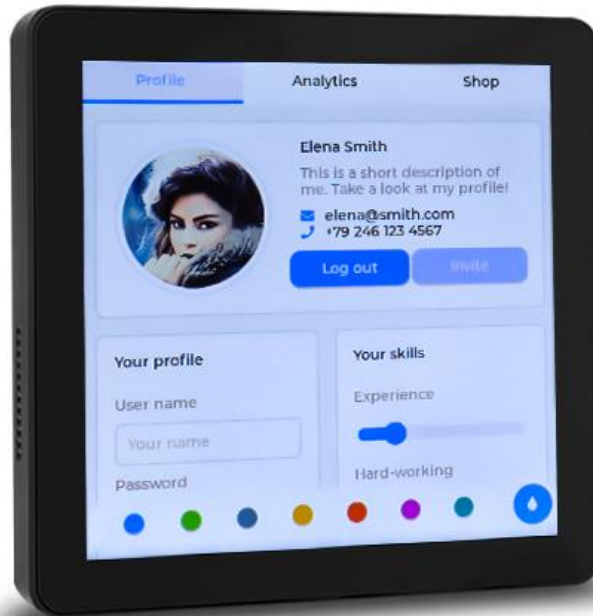




ESP32-S3-Touch-4-220V

4-inch Touch Display Development Board

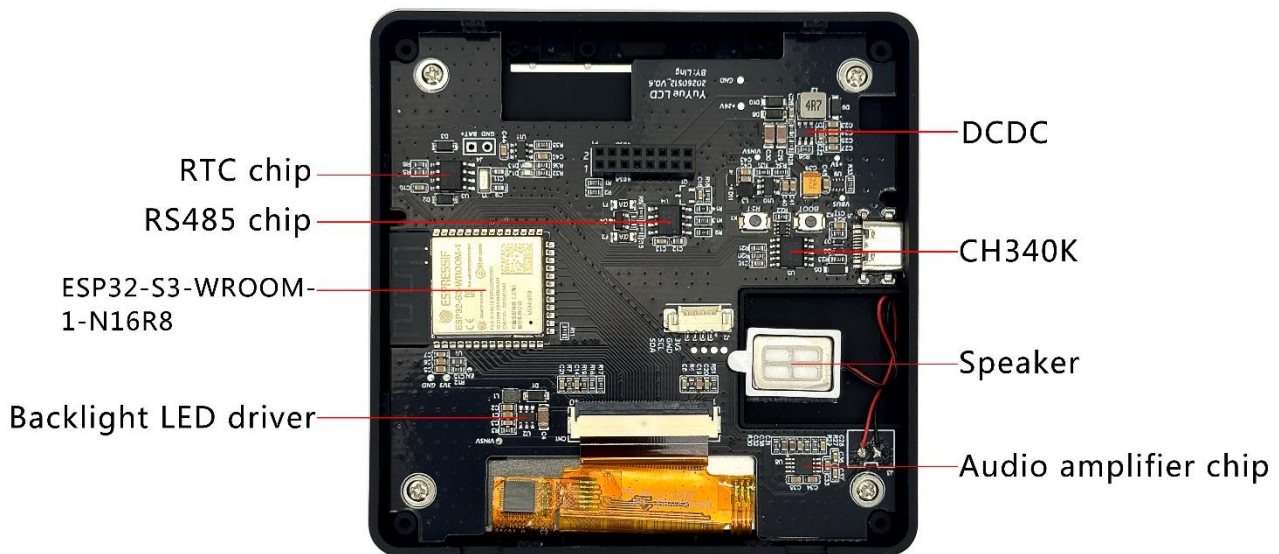
1. Product Overview

ESP32-S3-Touch-4-220V is an integrated smart touch control development board featuring ESP32-S3 Wi-Fi/BLE MCU, a 4-inch RGB capacitive touch display, 220V AC power input, 24V DC output and RS485 communication interface.

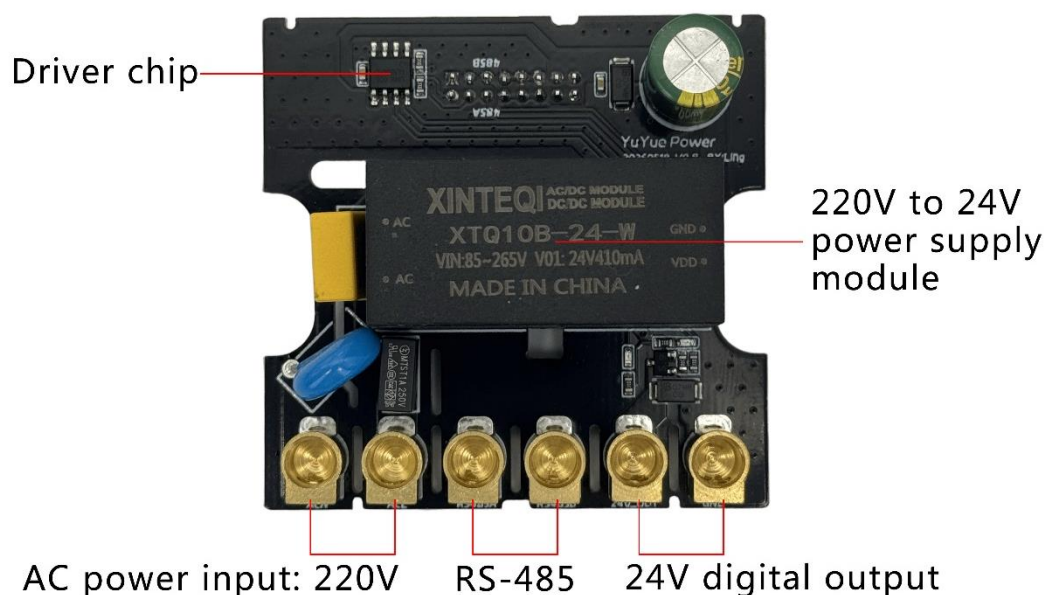
2. Hardware Configuration

- **ESP32-S3-WROOM-1-N16R8 Main Control Module:** 16MB Flash + 8MB PSRAM, supports WiFi and BLE 5.0
- **4-inch RGB Touch Display:** 480×480 resolution, used for graphical interface and human-machine interaction display
- **220V AC to 24V DC Power Supply Module:** Supports 85~265V AC input, outputs 24V DC
- **RS485 Communication Interface:** Supports industrial device communication
- **Speaker:** Provides audio output/playback functions
- **Power Management and Protection Circuitry**

3. Main Components



Front View: Structure and Main Function Callouts



Rear View: 220V Power Module and Interfaces

4. Interface Definition

Module	Function Description
ESP32-S3 Main Controller	High-performance WiFi + BLE MCU, supports AIoT application development
4-inch LCD Touch Screen	480×480 RGB display, supports touch interaction
220V Power Supply Module	AC85–265V input, 24V DC output
RS485 Interface	Industrial field communication interface
USB Type-C	Program download, debugging, and power supply
Speaker	Audio alert / sound notification

5. Interface Definition

Bottom Terminal Interface Definition:

- (1) AC Input: 220V AC power input
- (2) RS485A / RS485B: RS-485 communication interface
- (3) 24V Output: DC power output
- (4) GND: Power ground

6. Power Supply Instructions and Safety Precautions

- The device supports AC85–265V wide-range voltage input.
- The 220V AC input is a high-voltage area. Do not perform any operation while the power is on.
- Before first use, please verify that all wiring is correct to avoid short circuits that may damage the device.
- It is recommended that installation and debugging be carried out by personnel with experience in electrical operations.

7. Development and Usage Instructions

- (1) Connect to a computer via USB Type-C for program downloading and debugging.
- (2) Use the ESP-IDF or Arduino development environment for software development.
- (3) Develop LCD interface, touch interaction, and communication functions according to project requirements.

8. Application Scenarios

- Smart home control terminal
- Industrial equipment control panel
- Intelligent power control system
- IoT gateway
- Human-machine interface for automation equipment

9. Technical Support

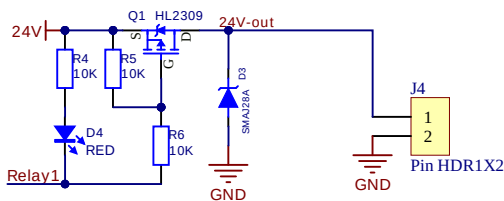
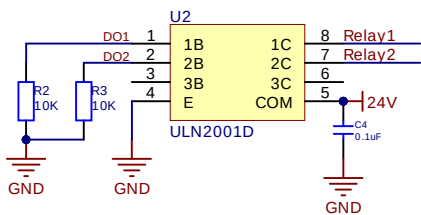
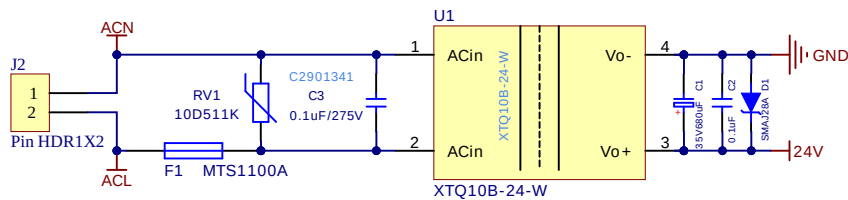
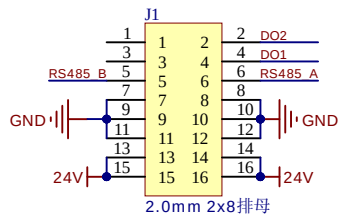
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10. Attachments

Schematic Diagram



		Sheet Title		Power	
		Project Title		Power	
Size	A4	Ver.	7	Author	Ling
Sheet	1	/	1	Data	20260416